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A Demographic-Aware Biased Matrix Factorization Approach for Cold-Start Course Recommendation

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ABSTRACT: Course recommender systems have moved from being mostly academic research topics to becoming significant practical tools for universities to guide students in selecting courses. Where they help students find their way through modular academic programs, inter-course dependencies, and catalogues of elective courses that are becoming more complex with each academic cycle. Collaborative filtering driven by biased matrix factorization (B-MF) is a popular approach in recommender systems due to its fast training, low latency and embedding representation that can be examined and recorded by academic regulators. But B-MF has one important limitation. If a new student is added to the system without any prior interaction history, the model cannot learn the student's preferences, thus making personalized recommendations not possible. In this paper, we propose a novel extension of B-MF, called Demographic-aware Biased Matrix Factorization (DB-MF). Which can mitigate the cold-start problem significantly and maintain the factorization framework that contributes to the effectiveness of B-MF. DB-MF improves the user latent vector by integrating a learned linear projection of the demographic attributes collected at the time of student enrollment, such as gender, age group, and previous educational attainment. The model is tested using the OULAD dataset. The whole process of evaluation is explained in detail with justifications.

KEYWORDS: Educational recommender systems; Demographic-aware; Biased matrix factorization; Collaborative filtering; Cold-start personalization

1 Introduction

Universities are using data analytics tools more and more in the academic advising process. These tools help students select the appropriate courses, plan their studies, and navigate ever more complex degree programs [1]. With more courses and more options for study offered by universities, students need help to make the right choices. The recommender system learns from the experiences of past students and uses this knowledge to recommend the best courses for current students [2]. Simple is the idea of recommending courses but building a good recommendation system is hard. The conditions at universities are very different from those on commercial platforms such as Netflix or Amazon, where most of the research on recommender systems has been focused on. Universities have specific challenges, such as limited student data and curriculum rules [3].

The number of courses offered by online stores is much larger than that of universities. In addition, each student takes only a handful of courses, so there is much less interaction data. This leads to sparse data and hence it is difficult to build accurate recommendation systems [4]. Student success data (e.g. grades and course completion) is only available once students have completed their courses. In addition, grading standards vary across departments and years, and universities need to make sure that recommendation systems are transparent, explainable, and fair to all groups of students. All of these factors make educational recommendation systems more challenging to develop than recommendation systems used in e-commerce [5]. Universities typically have small analytics teams that maintain many AI models for various departments and campuses. Such teams often lack the time and effort to keep complex deep-learning systems running, since deep-learning software moves fast and requires frequent updates [6]. Because universities have limited technical resources, limited data, and strict transparency rules. They want recommendation systems that are easy to train, that produce recommendations quickly, and that can explain why each recommendation was made.

Biased Matrix Factorization (B-MF) has been one of the most important recommendation algorithms for more than ten years. It is a robust and efficient approach which is the fundamental of many modern recommendation models, and still widely used in practical systems [7]. B-MF can quickly generate recommendations by learning compact representations of users and courses. It records important information such as the courses' popularity, the similarity between courses, and the latent features of users and items. Over the years, many researches have been conducted to improve the training and optimization techniques of B-MF, making it a reliable choice for practical recommendation systems. Various versions of matrix factorization have been implemented in the field of education [8]. These models have been successfully used to predict student performance and recommend appropriate courses. They are a favorite choice for university analytics teams because they are interpretable, reliable, and easy to maintain.

Matrix Factorization (MF) has been a successful recommendation approach for many years; however, it suffers from a fundamental cold-start limitation [9]. Because the model does not have any interaction history needed to build meaningful latent user representations, it cannot provide personalized recommendations for newly enrolled students. So, suggestions for these users are mainly derived from course popularity and average ratings, rather than personal preferences. The authors argue that universities already have access to useful demographic and admissions information such as age, gender, and previous educational achievement which has predictive signals about course choices, academic progression and expected performance. The principal research question is whether these existing demographic attributes can be integrated into the MF framework, in order to provide better recommendations for new students, while preserving the simplicity, efficiency and interpretability which make Matrix Factorization appealing [10].

Existing research includes very complex recommendation systems that combine deep learning with many types of data such as course descriptions, prerequisite relationships, student click behaviour and knowledge graphs [11,12]. These systems are often more accurate, but are harder to train, require more computing power and are harder to understand and maintain. Standard Matrix Factorization is often used as a baseline, but it cannot personalize recommendations for new students and instead recommends the most popular courses. Between simple MF and complex deep-learning models are lightweight methods that use additional information, such as

demographic features. Although these methods are commonly used in practice, they have rarely been evaluated and compared fairly under the same cold-start conditions using public educational datasets. Even when researchers compare recommendation models, they often do not perform enough evaluation. They rarely test whether the improvements are statistically significant, which demographic features contribute most, whether the model is fair to different groups of students, or whether it still works well for students who already have interaction history. Without such validation checks, it is difficult to determine whether the reported improvements are reliable and applicable in real-world settings. In this work, we study the possibility of incorporating demographic information (age, gender, and previous education) into B-MF by a simple linear transformation in order to improve the recommendations for the new students. We compare the proposed approach with other lightweight demographic-aware approaches in a fair, transparent, and statistically rigorous evaluation framework. The remainder of the paper is structured around three distinct research questions that are investigated experimentally and through data-driven evaluation:

- Can demographic-enhanced biased matrix factorization improve recommendations for cold-start students?
- How does DB-MF compare to other lightweight models that also utilize the same demographic attributes differently?
- Does demographic data help the model even after students have interaction history, and how does the model performance vary with different demographic features and student groups?

This paper is organized as follows: Section 2 reviews the related work; Section 3 presents the key notions; Section 4 introduces the problem setup; Section 5 provides the full mathematical formulation of the models; Section 6 describes the experiments; Section 7 discusses the results; and Section 8 concludes the paper.

2 Related Work

The explosive growth of user-item interactions and the emergence of digital platforms have motivated the design of recommendation algorithms. Their primary goal is to generate appropriate recommendations by identifying patterns in users' tastes and past behavior. The early recommender systems were based on simple heuristic methods like popularity-based ranking and neighborhood-based collaborative filtering. With the increasing complexity of the recommendation tasks, the research shifted towards model-based approaches such as matrix factorization, factorization machines and deep learning methods. Recently hybrid recommender systems have been developed by combining collaborative filtering with extra information such as demographic characteristics, item content, and contextual data. These methods are able to improve the accuracy of recommendations, especially in difficult situations such as the cold-start problem.

The authors of the paper [13] proposed a recommender system that uses various machine learning algorithms to predict suitable improvement actions based on course specifications, academic records and assessments of course learning outcomes to improve decision-making in outcome-based education. The recommendation task is defined as a multi-label multi-class classification problem and experimental results show the effectiveness of the proposed approach in the improvement of teaching strategies and students' learning experiences. A variety of

classification and transformation techniques are applied, increasing computational complexity, and may be a barrier to scalability in large educational settings.

Tianyuan *et al.* [14] propose an education recommender system that uses GPT to automatically extract knowledge concepts from course syllabi and build a knowledge graph to give personalized course recommendations. Experimental results show that the concepts generated by the GPT are highly consistent with the course content and significantly improve the accuracy and performance of the recommendation system. However, the method depends on the quality and consistency of the generated concepts and may require a significant amount of computational power. This is because the approach is based on GPT-generated knowledge graphs.

The paper [15] studies a hybrid recommender system that uses collaborative filtering, content-based filtering, and Association Rule Mining (ARM) to address the cold-start problem. The proposed approach improves accuracy and relevance of recommendations for new users and items by discovering hidden relationships in sparse datasets with association rules. Experimental results show that the recommendation quality is improved and user satisfaction is increased. However, the success of the model relies on the quality of the extracted association rules, which can be limited on fast-evolving or highly sparse datasets.

Zhang *et al.* [16] propose a hybrid recommendation framework to address the cold start problem of recommender systems. Their method considers the full cold start (CCS) and partial cold start (ICS) situations for new items. The model combines collaborative filtering and deep learning approaches. We use a deep neural network, SADE, to learn the content features of items. These are integrated into a modified timeSVD++ model for rating prediction. The model also considers the temporal dynamics of user preferences and item characteristics. Experimental results on the Netflix dataset show that the proposed method is effective compared to baseline models. More importantly, the authors also provide a flexible strategy that can be used to retrain and switch models to handle item transitions. However, it depends on high-quality content features and it requires high computational resources

Kumar *et al.* [17] introduces a personalized learning recommendation system to address the cold start problem of new users. It is important to understand the learner context and behavior for making effective recommendations. But such information is not available for new learners in cold start situations. Collecting this data via questionnaires may reduce user engagement. The authors tackle this problem by looking at learners' questions. Meta-information is extracted by the system to estimate learner needs and knowledge levels. Learner modeling and topic classification are performed using BiLSTM models and Ontology. Suitable learning materials are identified and ranked using context mapping and MCDM techniques. The experimental results show that the approach can effectively support the personalized recommendation for new learners.

The study [18] aims to enhance the conversion rates in e-commerce by utilising personalised recommendation systems. Typical collaborative filtering methods work well and are popular but suffer from sparsity and cold-start. In order to address these issues, the authors include additional data, such as user profiles and textual reviews. They proposed a hybrid model that combines self-supervised learning and collaborative filtering. In addition, BERT-DNN is used to learn the contextualised representation of item description. This helps to better understand user's preferences and item's features. The system uses ItemKNN and UserKNN to generate accurate recommendation. The proposed approach includes embedding generation, dimensionality

reduction, deep neural network construction and ensemble learning. The experimental results demonstrate that the proposed approach has been improved greatly.

3 Key Notions

3.1 Cold-Start Recommendation

Cold-start problems remain a central challenge for recommender systems, and a wide range of proposed solutions exists at both the algorithmic and system levels. In the item-cold setting, where no user interaction data is yet available to score new catalogue items, prior work has resorted to content-based embeddings, collective matrix factorization using auxiliary matrices, and local neighborhood-based models that transfer information from similar items [19]. The user-cold setting addressed in this work is especially fundamental, as personalization is based on a user embedding that cannot be estimated at scoring time. As shown in the figure 1. There are three main

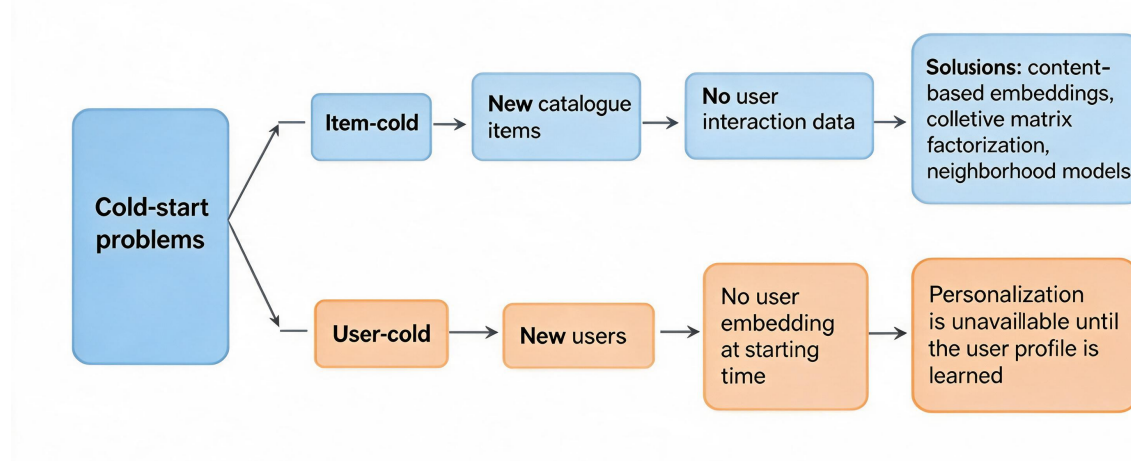


Figure 1: Schema for cold-start problems in recommender systems

families of solutions. Meta-learning methods train models to rapidly adapt from few interactions by learning an initialization that can be fine-tuned in a small number of gradient steps as soon as the first enrollments of a new user are observed. Hybrid models typically mix embedding lookups with text or image encoders in multi-branch architectures, using collaborative signals and content-based representations [20]. The attribute-conditioned models build a useful embedding from the registration-time features, essentially creating a cold-start latent factor that can be used in the standard scoring pipeline. DB-MF is a member of this third family. The main practical motivation is that institutional systems typically have access to demographic and academic-history fields at matriculation, and these are the only signals available prior to a learner registering for any course [21].

3.2 Educational Recommender Systems

Educational recommender systems have limitations that make them quite different from the consumer systems that dominate much of the recommendation literature. First, their results are noisy and lagged: grades can be published weeks or months after enrolment, intermediate signals such as virtual learning environment (VLE) clicks are sparse and irregular, and the same final grade

can reflect very different learning experiences across students [22]. Second, curriculum rules impose significant constraints on the set of feasible recommendations. Prerequisite chains, co-requisites, credit caps, and programme-specific regulations mean that recommendations have to be generated according to prerequisite logic, not from an unconstrained catalogue. Third, the stakeholder environment is complex and directly influences the design of the model [23]. The requirements of students, instructors, academic advisers, department heads and accreditation bodies overlap, leading to a multi-stakeholder objective that often demands transparent, interpretable inputs that can be audited and explained to non-technical audiences. These factors make interpretable hybrid approaches preferable to opaque deep models in general and in institutional settings where the catalogue size is typically small (tens to a few hundred items). As illustrated in the figure 2

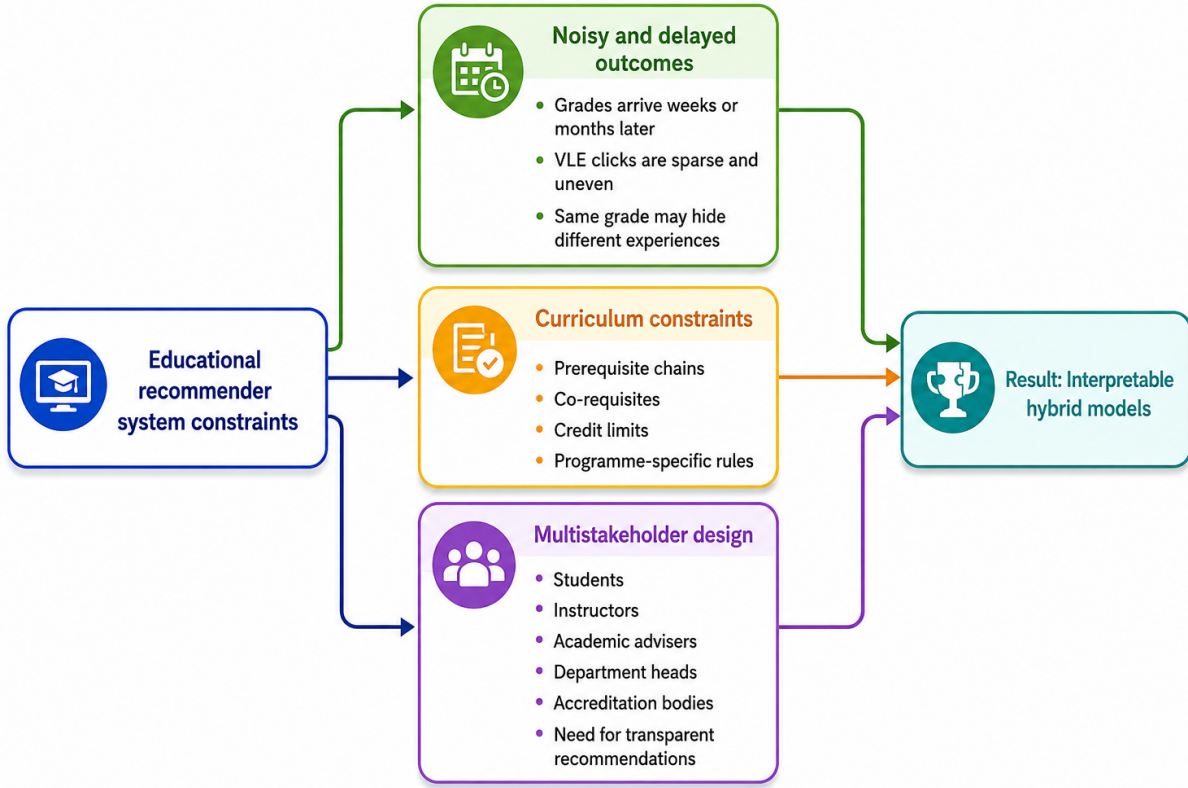


Figure 2: Design Constraints for Educational Recommender System

4 Problem Setup

In this section, we formally define the recommendation task, item and label construction, demographic encoding scheme, and cold-start evaluation protocol. All design choices are explicitly specified, as the previous section has shown that protocol-level decisions in educational recommendation benchmarks can have a larger impact on results than the differences between competing methods.

Let U be the population of students and I the list of items. We denote by $P \subseteq U \times I$ an implicit set of positive interactions. Each pair $(u, i) \in P$ indicates that student u passed (or got a distinction in) the module presentation represented by item i . For each student $u \in U$, a fixed-length

demographic feature vector $d_u \in \mathbb{R}^m$ is built from fields available at matriculation and encoded as described below. The training objective learns item latent factors and biases, as well as user latent factors and biases for students appearing in the training graph. For students appearing only at test time, there is no collaborative user factor available by construction. Hence, any personalization in the recommendations should be based on item-level parameters and demographic side information if the model allows. This operationally specifies the user-cold setting used throughout the paper.

Definition of item. The OULAD dataset [24] identifies enrollments by using a combination of two attributes: the module code (*code_module*) and the presentation code (*code_presentation*) which together specify a particular offering of a module during a particular academic period. The two attributes are combined to form a unique item identifier. There are two main arguments for this modeling choice. First, different presentations of the same module may vary in terms of start dates, pedagogical focus, and student cohorts and thus result in different interaction patterns; treating each presentation as a separate item preserves this heterogeneity. Second, combining all presentations of a module in one would reduce the item space to only 7 entities, making top-K evaluation metrics (e.g., $K = 5$) less informative due to the small catalog size. In contrast, items defined at the module-presentation level yield a catalog of $|I| = 22$ items. Although still relatively small, this level of granularity is sufficient to enable meaningful ranking evaluation and to distinguish performance across recommendation models.

Positive interactions. As is common in educational data mining applied to grade-oriented settings, an interaction is considered positive if the *final_result* field in the OULAD dataset contains either "Pass" or "Distinction"; enrollments with "Withdrawn" or "Fail" results are not included in the positive class, nor are records with missing or unspecified result values. This encoding defines relevance in terms of observed academic achievement rather than explicit user preference. While this is a simplifying assumption that has limitations discussed in Section 7, it has the practical advantage of being fully reproducible, using the standard OULAD dataset, without requiring additional annotation or subjective interpretation.

Table 1: Encoding of OULAD enrollment outcomes into implicit feedback.

final_result	Interaction Label	Treatment
Distinction	Positive ($r_{ui} = 1$)	Included
Pass	Positive ($r_{ui} = 1$)	Included
Fail	Negative ($r_{ui} = 0$)	Excluded from positive set
Withdrawn	Negative ($r_{ui} = 0$)	Excluded from positive set
Missing / Undefined	Undefined	Discarded

Demography of encoding: The feature vector d_u comprises of three demographic attributes from the OULAD *studentInfo.csv* table: the gender, age band and highest education level of the student. Each categorical feature is one-hot encoded with a vocabulary that is learned once from the training population and used as-is on cold-test students. This means that the index mapping of features is deterministic across experimental runs. The categories in each field are then sorted lexicographically to provide a stable encoding. Table 2 and figure 3 summarize the encoding

schema. The resultant feature dimensionality is $m = 10$ for two gender categories, three age bands and five education levels.

Table 2: Demographic encoding for d_u (concatenated one-hot vectors; $m = 10$ in the UCI extract used here).

Field	Cardinality	Categories (sorted lexicographically within each field)
Gender	2	F, M
Age band	3	0–35, 35–55, 55≤
Highest education	5	A Level or Equivalent; HE Qualification; Lower Than A Level; No Formal equals; Post Graduate Qualification

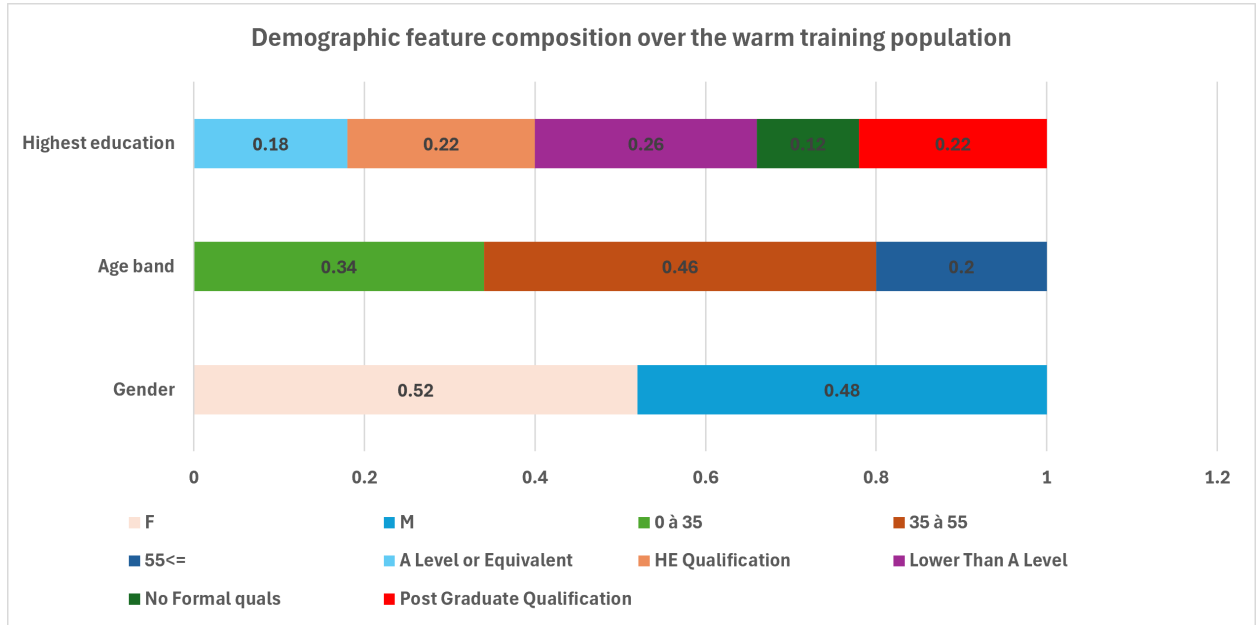


Figure 3: Demographic feature composition over the warm training population

Cold-test protocol: The cold-test cohort is created by a temporal split that is intended to simulate the operational context in which an institutional advising tool operates. Students are ordered by their earliest observed registration date in all enrollments, which records when they first registered for a module presentation. The first 80% of students in this ordering are assigned to the warm-user group, and all their positive interaction edges are used for training and validation. The last 20% of students are strictly the user-cold test cohort. None of the interactions with a cold-test student are part of the training or validation data. Thus, their user factors are not available at evaluation time. Any personalization in their recommendation lists must then come from item parameters and the demographic projection, if supported by the model. This protocol is meant to be different than a random split at the user level, which might leak temporal information by allowing the model to train on users that registered after some test users. Instead, it reflects a forward-looking setting, where a new student comes in and the system must make recommendations without having seen any enrollments. The proposed framework for constructing temporal cohorts is depicted in Figure 6.

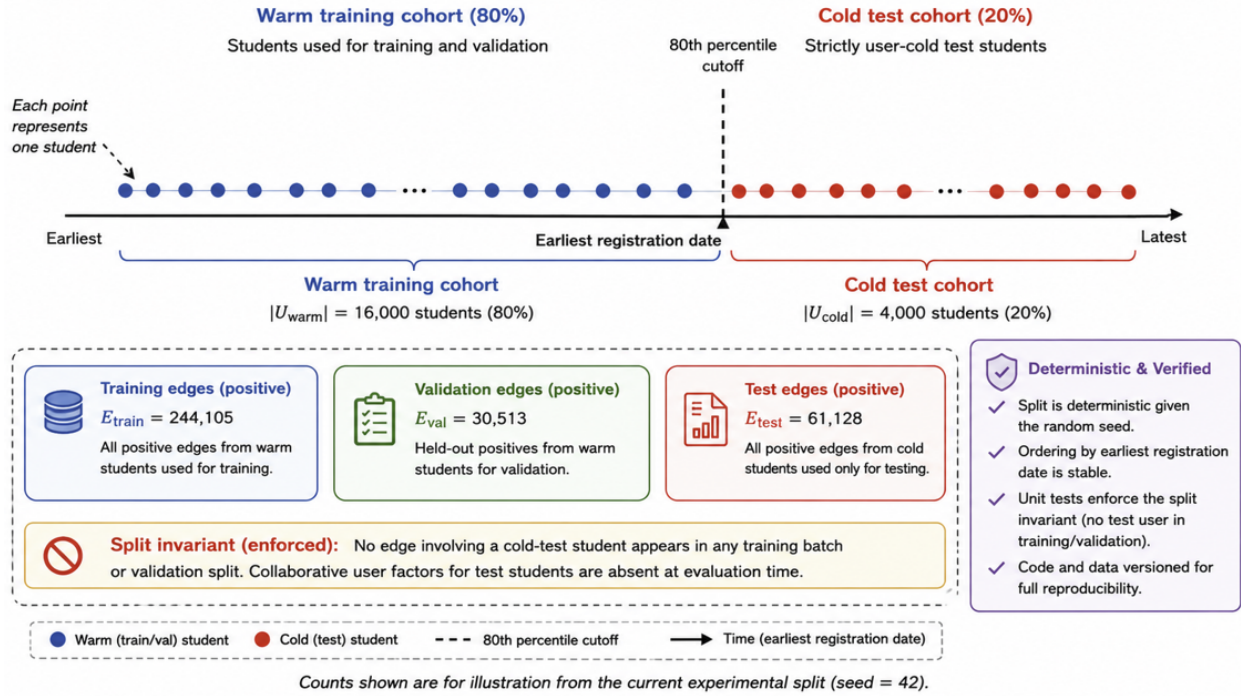


Figure 4: Proposed framework for constructing temporal cohorts

5 Models

The proposed approach is Demographic-aware Biased Matrix Factorization (DB-MF): a biased matrix factorization model that augments each user’s latent vector with a learned linear projection of demographic features available at matriculation, namely gender, age band, and highest education level. In the cold-start situation, this means the model can score a new student using demographics and item parameters, rather than falling back to popularity-only recommendations. In this section, we formalize the four learnable models that are evaluated in the empirical study along with the two baseline heuristics that act as calibration instruments. All four learnable models share the same pairwise BPR-style training loss, the same AdamW optimizer, and the same negative sampling regimen, differing only in how the user-side information is parameterized and how cold-start scoring is performed.

5.1 Biased Matrix Factorization (B-MF)

Biased Matrix Factorization (B-MF) is a model for recommender systems that estimates the rating of a user as a combination of a global average, a user bias, an item bias, and the interaction of user and item latent factors [25]. The typical form is:

$$\hat{r}_{ui} = \beta + b_u + b_i + \langle \mathbf{p}_u, \mathbf{q}_i \rangle, \quad (1)$$

where $\beta \in \mathbb{R}$ is the overall bias term, $b_u \in \mathbb{R}$ and $b_i \in \mathbb{R}$ are user and item bias terms, and $\mathbf{p}_u, \mathbf{q}_i \in \mathbb{R}^f$ are the user and item latent factor vectors, respectively. For users who appear only in the test set (i.e., new users with no interaction history), the parameters associated with the user $\mathbf{p}_u$

and b_u are unavailable, as they were not part of the training set. The cold B-MF score then becomes $\hat{r}_{ui}^{\text{cold}} = \beta + b_i$, a score that only depends on the overall bias term and the item bias that is learned.

5.2 Proposed approach: DB-MF

DB-MF stands for Demographic-Aware Biased MF, and it is a matrix factorization recommender model. It is an extension of biased matrix factorization, which includes demographic information (e.g., age, gender, or other user attributes) in the prediction process. In our approach, we augment the user latent vector with a learned linear projection of the demographic feature vector. The augmented user factor is defined as:

$$\tilde{\mathbf{p}}_u = \mathbf{p}_u + \mathbf{W}\mathbf{d}_u, \quad \mathbf{W} \in \mathbb{R}^{f \times m}, \quad (2)$$

and the score is computed via the same inner-product interface as B-MF:

$$\hat{r}_{ui} = \beta + b_u + b_i + \langle \tilde{\mathbf{p}}_u, \mathbf{q}_i \rangle. \quad (3)$$

During training, the demographic vector $\mathbf{d}_u$ is available for every warm user, and the projection matrix $\mathbf{W}$ is learned jointly with the embeddings, biases, and all other learnable parameters under the same BPR objective.

In the cold setting at test time, we do not have the interaction-derived user vector $\mathbf{p}_u$ and user bias b_u , thus we set them to zero. Thus, the DB-MF score for the cold-start case only depends on the item parameters and the demographic projection:

$$\hat{r}_{ui}^{\text{cold}} = \beta + b_i + \langle \mathbf{W}\mathbf{d}_u, \mathbf{q}_i \rangle. \quad (4)$$

The geometric interpretation of this construction is simple. $\mathbf{W}$ is a deterministic generator of the embeddings: it maps a student's one-hot demographic encoding into the same f -dimensional latent space as the item factors, and subsequently uses the resulting embedding in the standard dot product scoring function. It doesn't add any model components, loss terms or inference branches. The parameter overhead over B-MF is exactly $f \cdot m$, which is 200 additional parameters at $f = 20$ and $m = 10$. Figure 5 shows the architectural difference between B-MF and DB-MF.

The score for B-MF (Bayesian Matrix Factorization) is computed as a dot product between user and item latent factors. DB-MF incorporates a learned linear projection $\mathbf{W}\mathbf{d}_u$ of the demographic feature vector and combines it with the interaction-driven user factor $\mathbf{p}_u$ prior to the dot product. The global mean and bias terms are present in both models, but have been omitted from the diagram for visual clarity. The Concat-MF and FM architectures described in the text follow the same input-transformation-dot-product pipeline but differ in how they parameterize the demographic features.

5.3 Feature-Concatenated MF (Concat-MF)

Concat-MF is another way to bring demographic information into the matrix factorization framework [26]. Instead of projecting demographics into the latent space with a learned projection

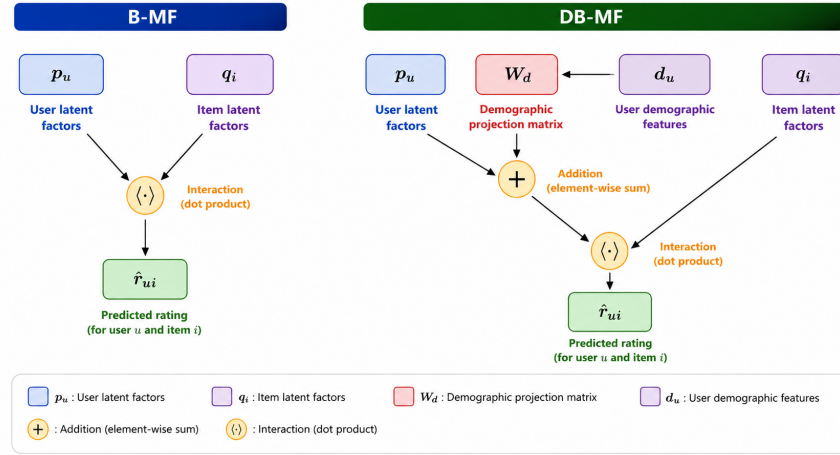


Figure 5: Architectural comparison of B-MF and DB-MF.

matrix, it constructs an extended user vector by concatenating the interaction-driven user factor and the raw demographic features, $[\mathbf{p}_u; \mathbf{d}_u] \in \mathbb{R}^{f+m}$. To match this higher dimensional representation, the item factor is also expanded to $\mathbf{q}_i \in \mathbb{R}^{f+m}$. Then the predicted score is calculated as $\hat{r}_{ui} = \beta + b_u + b_i + \langle [\mathbf{p}_u; \mathbf{d}_u], \mathbf{q}_i \rangle$ and the model is trained in the same way as B-MF and DB-MF with BPR objective. For a cold user $\mathbf{p}_u = 0$ and $b_u = 0$; the prediction is only based on the interaction of the demographic part of the extended user vector with the related item dimensions. Thus, Concat-MF has the same information available at scoring time as DB-MF: both models have the same demographic features for cold users and the same collaborative signal for warm users, but they encode that information via different parameterizations. The gap between DB-MF and Concat-MF demonstrates the impact of the linear projection with a reduced dimension: Concat-MF enables demographic features to interact directly with item factors in the full joint embedding space, while DB-MF compresses the demographic signal via $\mathbf{W}$ prior to the dot product.

5.4 Factorization Machine (FM)

The factorization machine is implemented as a dense second order model over a sparse indicator feature representation [27]. For each training instance, five feature indices are active: one for the warm user id, one for the item id and one for each of the three active demographic one-hot dimensions. We decompose the FM score into a global bias, first-order feature weights and the sum of pairwise inner products of the low-rank embedding vectors of all active feature pairs. The factorization rank is set to 20, which is the same as the latent dimensionality f in the MF variants. For cold scoring, we drop the user-identifier feature from the active set, so the prediction is based only on the item embedding, the demographic feature embeddings and all pairwise interactions between them. In this sense, FM is more expressive than DB-MF or Concat-MF, as demographic features interact with each other and with the item embedding directly through multiple pairwise terms, instead of being passed through a single projection matrix.

5.5 Demographic Popularity (Demog-POP)

Demog-POP is not a learned model, but a simple cohort-based heuristic included as a calibration baseline. For each cold-test user it finds the subset of warm-training students who

match that user on the full demographic tuple of gender, age band and highest education. Items are then ranked by the number of positive interactions in the matched subgroup [28]. For rare demographic combinations, there might not be any training student that matches the cold user on all three demographic dimensions. In such cases, the heuristic falls back on ranking the items according to their overall popularity in the training set. The point of adding Demog-POP is to understand how much cold-start signal is captured by raw cohort-level statistics alone, with no latent-factor modeling whatsoever. If a learned model cannot beat this heuristic on cold-start metrics, then it is not extracting any meaningful value from its learned parameters.

6 Experiments

This section presents the complete empirical evaluation of the proposed method. We first report the dataset statistics and define the evaluation metrics. Then we present the main cold-start results. Next, we do a demographic ablation experiments to quantify the contribution of each demographic component. Finally, we report subgroup-level metrics to look at the performance across user segments.

6.1 Dataset Overview

The UCI redistribution of OULAD (dataset ID 349) is converted to a set of positive interaction edges for training, as per the procedure in Section 3. This process produces an interaction graph with 15,385 positive enrollments by 14,581 unique students over $|I| = 22$ items at the module-presentation level. The data is split temporally into 11,664 warm users for training and 2,917 cold users for testing. For the warm user interactions, we consider 11,138 edges in training and 1,238 edges in validation for the early stopping step. Figure 2 provides a visual summary of these statistics.

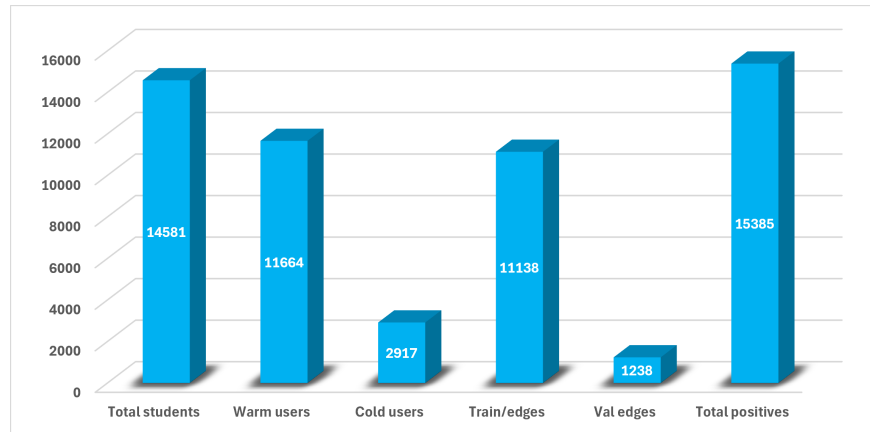


Figure 6: OULAD dataset: Pipeline counts after temporal split of students

6.2 Evaluation Metrics

For each cold-test user, the relevant items are defined as the subset of the 22 catalogue items which the user passed, i.e., those with a Pass or Distinction outcome in the held-out cold cohort. The quality of the recommendation is then evaluated at $K=5$ using three measures, namely, Precision@5, Recall@5, and Normalized Discounted Cumulative Gain at 5 (NDCG@5).

1) Precision@K:

Precision@K measures how many of the top K recommendations are correct using the following equation 5.

$$\text{Precision@K} = \frac{1}{K} \sum_{i=1}^K \text{rel}_i \quad (5)$$

Where:

- K : the number of top recommended items considered. In your case, $K = 5$.
- i : the rank position in the recommendation list, from 1 to K .
- rel_i : the relevance of the item at rank i . For binary relevance, rel_i is usually:

$$\text{rel}_i = \begin{cases} 1 & \text{if the item at rank } i \text{ is relevant} \\ 0 & \text{otherwise} \end{cases}$$

2) Recall@K:

Recall@K measures how much of the user's relevant set was recovered in the top K using the following equation 6.

$$\text{Recall@K} = \frac{\sum_{i=1}^K \text{rel}_i}{|\mathcal{R}|} \quad (6)$$

Where:

- $\sum_{i=1}^K \text{rel}_i$: the number of relevant items found in the top K recommendations.
- $|\mathcal{R}|$: the total number of relevant items for that user.
- $\mathcal{R}$: the set of all relevant items for the user.

3) NDCG@K:

NDCG@5 (Normalized Discounted Cumulative Gain) is a more sensitive metric that measures the relevance and the quality of the ranking of top-5 recommendations by taking into account the position of relevant items in the ranked list. This metric is computed using the following equation 7.

$$\text{NDCG@K} = \frac{\text{DCG@K}}{\text{IDCG@K}} \quad (7)$$

The *Discounted Cumulative Gain at K* (DCG@K) measures the quality of a ranked recommendation list by assigning greater importance to relevant items appearing near the top of the ranking. It is defined as

$$\text{DCG@K} = \sum_{i=1}^K \frac{2^{\text{rel}_i} - 1}{\log_2(i + 1)}, \quad (8)$$

where K is the evaluation cut-off (e.g., $K = 5$), and rel_i denotes the relevance score of the item ranked at position i .

For binary relevance, where $\text{rel}_i \in \{0, 1\}$, the expression simplifies to

$$\text{DCG@K} = \sum_{i=1}^K \frac{\text{rel}_i}{\log_2(i+1)}, \quad (9)$$

where $\text{rel}_i = 1$ if the item at rank i is relevant and $\text{rel}_i = 0$ otherwise.

The *Ideal Discounted Cumulative Gain at K* (IDCG@K) represents the maximum possible discounted cumulative gain and is obtained by ordering all relevant items before non-relevant items. It is defined as

$$\text{IDCG@K} = \sum_{i=1}^K \frac{2^{\text{rel}_i^*} - 1}{\log_2(i+1)}, \quad (10)$$

where rel_i^* denotes the relevance score of the item at position i in the ideal ranking.

For binary relevance, where $\text{rel}_i^* \in \{0, 1\}$, the expression simplifies to

$$\text{IDCG@K} = \sum_{i=1}^K \frac{\text{rel}_i^*}{\log_2(i+1)}, \quad (11)$$

where $\text{rel}_i^* = 1$ if the item at rank i is relevant in the ideal ranking and $\text{rel}_i^* = 0$ otherwise.

All three metrics are calculated under binary relevance (an item is either relevant or not) and macro-averaged over users who have at least one relevant item in the cold cohort. Macro averaging is preferred over micro averaging, as it assigns equal weight to each user, which is consistent with the per-student fairness perspective that is commonly adopted in educational recommendation research and with the reporting practices of similar studies.

6.3 Main Results Under Cold-Start Conditions

The macro-averaged cold-test results for each of the five recommenders are presented in Table 3 as means $\pm$ standard deviations for each user. To make comparison easier, Figure 7 presents these findings as a bar chart.

Table 3: Cold-test cohort metrics: Means $\pm$ standard deviations across the 2,917 cold users.

Model	Precision@5	Recall@5	NDCG@5
POP	0.0660 ± 0.0951	0.3186 ± 0.4611	0.1764 ± 0.2707
B-MF (cold)	0.0660 ± 0.0951	0.3186 ± 0.4611	0.1930 ± 0.3072
FM (cold)	0.0850 ± 0.1000	0.4125 ± 0.4876	0.2679 ± 0.3492
Concat-MF (cold)	0.0810 ± 0.0996	0.3923 ± 0.4837	0.2539 ± 0.3411
DB-MF (cold)	0.0922 ± 0.1015	0.4483 ± 0.4933	0.2813 ± 0.3441
Demog-POP	0.0894 ± 0.1006	0.4343 ± 0.4907	0.2772 ± 0.3466

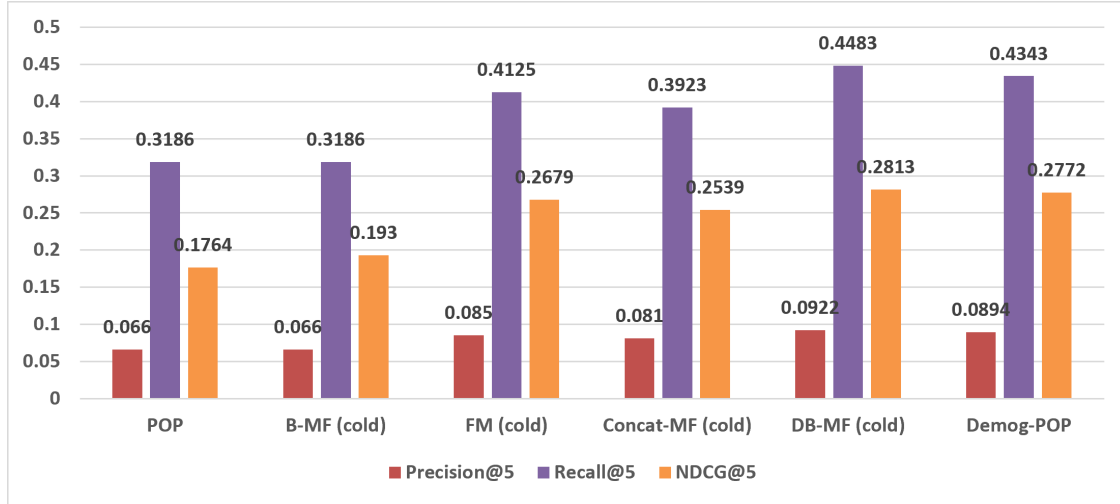


Figure 7: Cold-test ranking metrics across all six recommenders.

The results presented in Figure 7 and Table 3 lead to three substantive observations. **First, demographic information is shown to significantly improve personalization for the cold-start problem.** DB-MF outperforms all six evaluated recommenders by 0.0922 in Precision@5, 0.4483 in Recall@5, and 0.2813 in NDCG@5. The improvement is consistent across all three metrics, capturing both the relevance of the recommended items (Precision and Recall) and the quality of their ranking (NDCG). Therefore, the observed gain is not driven by a single metric or a narrow dimension of recommendation performance.

Second, **the functional form of demographic absorption matters.** DB-MF yields better results than Concat-MF on all metrics, with Precision@5 of 0.0922 versus 0.0810, Recall@5 of 0.4483 versus 0.3923, and NDCG@5 of 0.2813 versus 0.2539. This pattern implies that the dimensionality-reducing linear projection W is preferred over direct feature concatenation in Concat-MF given the current catalogue size and training budget: Combining demographic information with item factors in the f -dimensional latent space before combining them seems to provide regularization and prevent overfitting, which leads to improvements in ranking quality.

Third, **Demographic data offers substantial improvement in popularity-based and matrix-factorization-based cold-start recommendation.** Demog-POP outperforms the global POP baseline by achieving Recall@5 from 0.3186 to 0.4343, Precision@5 from 0.0660 to 0.0894, and NDCG@5 from 0.1764 to 0.2772. These correspond to relative gains of 36.3%, 35.5%, and 57.1%, respectively. Thus, demographic-group popularity results in much more relevant and better-ranked recommendations than global popularity only. A similar pattern applies to matrix factorization. In Precision@5 and Recall@5, B-MF (cold) performs exactly like POP, suggesting that collaborative latent factors by themselves are unable to consistently tailor recommendations for users in the absence of adequate interaction history. Conversely, a DB-MF (cold) with demographic information increases the Precision@5 to 0.0922, Recall@5 to 0.4483, and NDCG@5 to 0.2813. The improvements over B-MF are 0.0262 in Precision@5, 0.1297 in Recall@5, and 0.0883 in NDCG@5.

6.4 Demographic Ablations

We conducted a series of ablation experiments to determine which demographic attributes are responsible for the cold-start improvements and to study their interactions. In each experiment, we zero out certain demographic fields during training and cold-test inference. The resulting DB-MF cold-start metrics are shown in Table 4 and a visual comparison is provided in Figure 8.

Table 4: DB-MF cold-start performance under demographic feature masking.

Active demographic fields	Precision@5	NDCG@5
Gender only	0.0828 ± 0.0999	0.2584 ± 0.3387
Age band only	0.0522 ± 0.0894	0.1528 ± 0.2820
Education only	0.0521 ± 0.0886	0.1609 ± 0.2986
Gender + age band	0.0834 ± 0.0999	0.2594 ± 0.3381
Gender + education	0.0882 ± 0.1007	0.2746 ± 0.3480
Age band + education	0.0557 ± 0.0906	0.1675 ± 0.2976
All fields	0.0850 ± 0.1000	0.2679 ± 0.3492

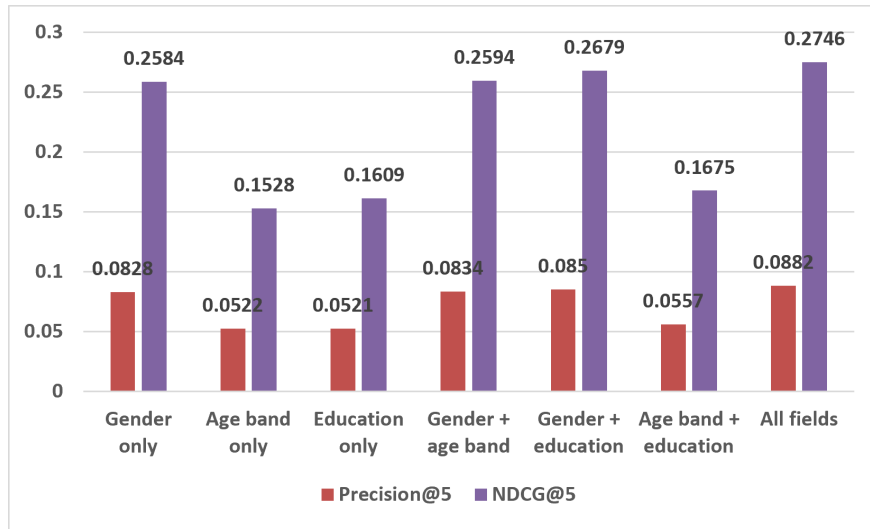


Figure 8: Demographic ablation profile for DB-MF.

Two patterns are clear. First, **in this pipeline, gender is the most prevalent demographic characteristic**. The gender-only configuration recovers the large majority of the full-model gain over B-MF cold, with Precision@5 of 0.0828 and NDCG@5 of 0.2584. Age band and education alone, on the other hand, perform close to the informationless baseline, indicating that neither factor carries enough signal on its own to be helpful for cold-start ranking on this catalog.

Second, **the attributes interact in non-trivial ways**. The gender–education combination has a Precision@5 of 0.0882, better than the all-fields configuration of 0.0850, but not as good as the all-fields NDCG@5 (0.2746 versus 0.2679). Age band appears to contribute little marginal signal beyond what gender already captures in this dataset, as the gender-age combination yields metrics that are almost identical to gender alone. The age + education combination (Precision@5 of 0.0557) is only a slight improvement over the single-attribute configurations for age and education,

confirming the main contributor to the demographic signal, gender, with education providing a secondary, interacting role.

6.5 Subgroup Metrics

In order to ensure algorithmic fairness in educational recommender systems, performance must be assessed both overall and among student subgroups that are safeguarded by institutional governance frameworks. Figure 9 shows a visual comparison of the macro-averaged DB-MF cold-test metrics broken down by gender, age band, and educational attainment, as shown in Table 5.

Table 5: Cold-start DB-MF performance mean $\pm$ standard deviation across gender, age-band, and educational-attainment subgroups.

Subgroup	n	Precision@5	NDCG@5
Male (M)	1502	0.0754 ± 0.0991	0.2372 ± 0.3418
Female (F)	1415	0.0951 ± 0.0999	0.3005 ± 0.3542
Youngest age band (0–35)	1870	0.0829 ± 0.1003	0.2591 ± 0.3455
Oldest age band ($55 \leq$)	16	0.0500 ± 0.0894	0.1808 ± 0.3338
Lower-half education	1835	0.0851 ± 0.1004	0.2641 ± 0.3433
Upper-half education	1082	0.0847 ± 0.0992	0.2743 ± 0.3592

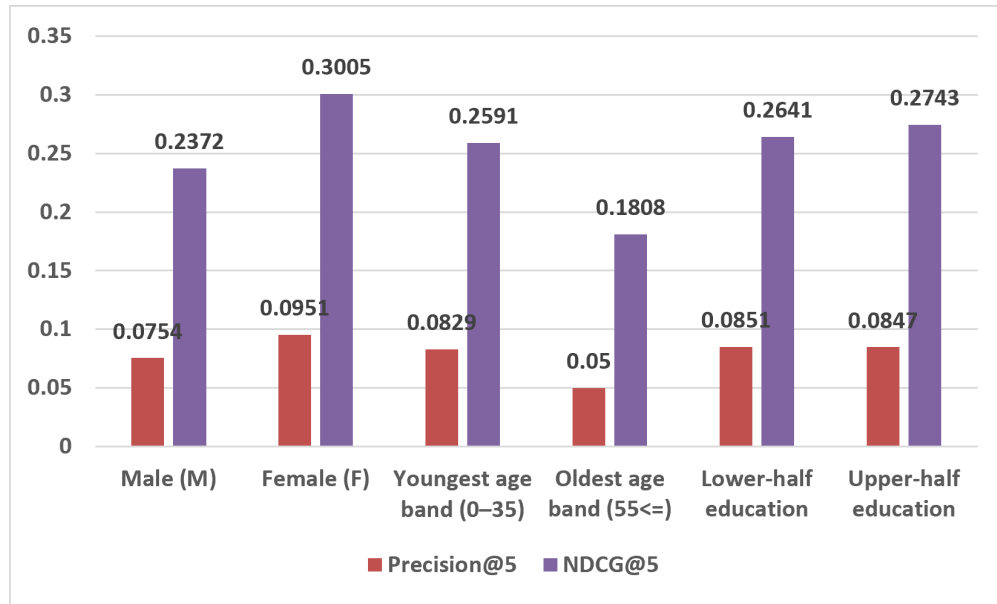


Figure 9: DB-MF cold-start performance by student subgroup.

A performance gap between gender groups is revealed by the disaggregation: female-identifying students receive higher mean NDCG@5 (0.3005 versus 0.2372) and higher mean Precision@5 (0.0951 versus 0.0754) than male-identifying students. The metrics for the largest component of the cold cohort, the youngest age group (0–35) with $n=1,870$ students, are similar to the averages of the total population. Since there are only 16 students in the cold cohort in the oldest age band ($55 \leq$), its subgroup estimates should be considered illustrative rather

than statistically stable. With a small NDCG advantage for the upper-half group (0.2743 versus 0.2641), which may indicate variations in the distribution of pertinent items among these subgroups, the education-based split shows generally comparable performance between the lower-half and upper-half attainment groups.

These subgroup differences should not be taken as evidence that DB-MF is inherently biased for or against a particular demographic group. Instead, they may be due to the interactions of the data generating process, temporal split, attribute encoding and recommendation task. But such disparities must be identified and documented under institutional governance frameworks prior to the implementation of a student-facing model. We therefore report them as part of a transparent performance audit and not as evidence of corrective intervention or normative judgment.

7 Discussion

Our results indicate that incorporating demographic information in a biased matrix factorization model enhances cold-start course recommendation significantly. DB-MF learns a linear projection from gender, age and education into the latent factor space, and results in a significant improvement in ranking quality, improving NDCG@5 by around 39% over the cold-start B-MF baseline. The improvements are consistent across a number of evaluation metrics and demographic ablation studies, while warm-start performance is also maintained or improved. These results show that demographic data available at registration of students are an effective replacement for lack of historical interaction data.

DB-MF has several practical advantages for real world deployment. It keeps the simple matrix factorization scoring process that leads to faster inference latency and higher computational efficiency. Also, its single demographic projection matrix gives a better understanding of the influence of demographic characteristics on recommendations. DB-MF is also more parameter efficient, with only a small fixed number of additional parameters independent of the number of users or items, which makes it easier to deploy and maintain in different educational systems.

Another important aspect of this work is reproducibility. All reported results can be reproduced with the provided code and configuration, as the entire experimental pipeline is deterministic and fully documented. For reproducible and reliable replication, relevant implementation details such as the temporal split of data, preprocessing of demographic data, bootstrap settings, software versions and random seeds are documented. The reproducibility of the framework enables future researchers to verify, compare and extend the proposed approach with confidence.

8 Conclusion

In this paper, we proposed and evaluated Demographic-Aware Biased Matrix Factorization (DB-MF), a parsimonious extension of biased matrix factorization that incorporates matriculation-time demographic attributes through a learned linear projection into the user latent representation. We benchmarked DB-MF against engineering-comparable alternatives – factors-concatenated matrix factorisation and a demographic-popularity heuristic – using the OULAD module-presentation catalogue under a strict user-cold evaluation protocol. Every protocol decision was explicitly documented and unit tests were performed to ensure the integrity of the implementation and evaluation pipeline.

The empirical results support three separate internally consistent conclusions. First, DB-MF outperforms the cold-scored biased MF and the global popularity baseline on all three metrics, which indicates that the demographic projection improves the cold-start ranking quality across metrics, random seeds and test-cohort subsets. Second, demographic conditioning is complementary to collaborative information, not a substitute. DB-MF also boosts warm-start performance over B-MF when interaction histories are available, indicating that the demographic features provide predictive value beyond the user representation learned from interactions. Third, DB-MF achieves the best overall cold-start performance on this compact catalogue, while the demographic-popularity heuristic is comparable to its results without learned parameters. This result indicates that a large part of the improvement in the cold-start scenario is due to the predictive power of demographic attributes per se, and not only due to the specific projection procedure used by DB-MF. However, DB-MF still provides benefits by incorporating this demographic signal with collaborative structure, while providing low inference latency, an interpretable latent-space formulation, and limited parameter overhead, thus favorably positioning it on the simplicity–performance frontier.

In future work, we will extend DB-MF in several directions to further improve recommendation quality and practical applicability. We will explore the application of sequential models to incorporate temporal student interaction data from the virtual learning environment, the integration of course content and prerequisite knowledge in hybrid recommendation systems, and the formulation of constraint-aware course planning, accounting for programme requirements and credit constraints. Additionally, we aim to explore uncertainty estimation for confidence-aware recommendations and conduct thorough fairness evaluations, including counterfactual and intersectional analysis, to foster equitable recommendations for demographic groups, all while upholding the transparent and reproducible framework established in this work.

Supplementary information

No supplementary information is associated with this article.

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Declarations

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Competing interests

The authors declare no competing interests.

Consent to participate

Not applicable.

Ethics statement

Not applicable.

Consent to publish declaration

Not applicable.

Data availability

The dataset used in this study are publicly available. The WSN-DS dataset can be accessed through <https://doi.org/10.1155/2016/4731953>.

Materials availability

Not applicable.

Code availability

The code used for model training and evaluation is available from the corresponding author upon reasonable request.

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